

Poster: GEM-PLI Side Channel in GPON: Inferring Encrypted Streaming Traffic

Hyunjun An
Heungdeok High School
hjahn@anhyunjun.com

Abstract—Gigabit Passive Optical Networks (GPON), an ITU-T standardized technology adopted in FTTH networks, supports downstream payload encryption to protect subscribers' traffic. Unfortunately, encryption does not cover all access-layer metadata. This poster presents a preliminary analysis demonstrating that the Payload Length Indicator (PLI), which remains in plaintext in GEM headers, can be exploited as a side channel to infer streaming activity. Using a GPON downstream simulator, We demonstrate that PLI traces are highly consistent across repeated runs of the same video and suggest that a passive attacker on the same optical splitter as the victim could potentially infer a victim's streaming activity (e.g., Youtube Video Identification).

I. INTRODUCTION AND MOTIVATION

As fiber internet has become dominant in network infrastructures, GPON—one of the passive optical network (PON) technologies—is adopted by many ISPs to provide fiber-to-the-home (FTTH) internet. Since GPON is a popular PON technology, it is important to ensure traffic confidentiality. GPON supports AES-based downstream payload encryption at the GPON Encapsulation Method (GEM) layer. However, some fields in the GEM header remain plaintext such as the Payload Length Indicator (PLI) field used for multiplexing and the Port-ID field used for mapping subscribers' Optical Network Terminals/Units (ONTs/ONUs).

Such plaintext fields may appear innocuous. However, recent studies have demonstrated that such metadata can be exploited to infer a victim's activity (e.g., identifying a victim's web-based video streaming watching activity) [4] [5] [6].

While prior studies primarily focused on upper-layer or cellular traffic analysis [4] [5], we researched whether the PLI field in GEM headers can be exploited to fingerprint a victim's streaming activity.

In this poster, We developed a software-based GPON downstream simulator to analyze the statistical properties of PLI traces. Our preliminary results indicate that Youtube streaming traces are highly consistent over time and distinctive at the GEM layer, suggesting that value of the PLI field can be exploited to infer victim's video streaming.

II. BACKGROUND

GPON framing. In GPON, the Optical Line Terminal (OLT) broadcasts downstream traffic over a passive optical splitter to multiple Optical Network Units (ONUs) (typically called "modem"). Downstream transmission is organized into fixed-duration GTC frames (125 μ s), each composed of a

physical control block downstream (PCBd) followed by a downstream GTC payload region. [1]

GEM and plaintext metadata. Upper layer traffic (e.g., Ethernet frame) is carried in the Payload Field in GEM frames. A GEM frame starts with a 5-byte GEM header containing a 12-bit PLI, a 12-bit Port-ID, a 3-bit Payload Type Indicator (PTI), and a header error control (HEC). The 12-bit PLI allows payload fragments up to 4095 bytes, so larger user frames may be fragmented. PTI indicates whether the payload is a non-terminal fragment or the end of a fragmented user frame. [1]

PLI plaintext length even under encryption. In GPON, downstream user traffic is carried as GEM frames multiplexed in the GTC payload [1]. Each GEM header includes a 12-bit payload length indicator (PLI) that specifies the payload length L in bytes (up to 4095 bytes), and larger user frames are fragmented accordingly [1]. Downstream GEM payloads are often encrypted using AES in CTR mode (XOR-based stream encryption). Therefore, even when payload bytes are encrypted, an attacker can observe the plaintext payload length per fragment via PLI.

Video Identification. HTTP Adaptive Streaming (HAS) is the popular video streaming model – which is generally adopted by video streaming platforms (e.g., Youtube) – delivers video by letting the client periodically request short media segments at a bitrate chosen by an adaptive bitrate (ABR) algorithm. This request–response process forms characteristic bursty download patterns and bitrate-dependent segment sizes, which can be exploited for traffic inference. [4]

III. THREAT MODEL

We consider an attacker who is on the same optical splitter as the victim and has the capability to obtain the victim's Port-ID, which is crucial for targeting specific GEM frames.

- **1. Premise** The attacker has an optical capture device (e.g., FPGA with optical TAP, modified ONU that can receive raw GTC frames) connected to splitter and passively records downstream from OLT. the attacker does not need to register their device with OLT; so even if they use receive-only optical tap, they can entirely record downstream records without transmitting.
- **2. Collect GEMs** Then, attacker extracts GTC frames from downstream traffic and parses GEM frames.
- **3. Convert GEM PLI to dataset** Then, the attacker filters GEM frames for a victim's Port-ID and records PLI values with timestamps.

- **4. Dataset format** Each record includes `t_start_us`, `gem_id`, `port_id`, and `pli`.
- **5. Classification via database matching** The attacker matches the captured PLI time series to a pre-collected database of PLI traces (built from known videos) and selects the closest match to infer the victim's video.

IV. TRACE GENERATION

To demonstrate the threat model in our experimental environment, we developed a PCAP-driven, software-based GPON downstream trace generator [3] that follows key ITU-T G.984.3 TC-layer parameters relevant to PLI (125 μ s framing and GEM header formatting). From a PCAP of encrypted video traffic, we (i) map Ethernet frames into GEM payloads (fragmenting to satisfy the 12-bit PLI limit), (ii) generate plaintext GEM headers (PLI/Port-ID/PTI/HEC), and (iii) pack the GEM stream into fixed-length 125 μ s downstream frames consistent with the configured line rate. The attacker signal is either the GEM-level PLI sequence or the per-frame occupancy series. Large Ethernet packets may generate multiple consecutive GEM fragments; thus, PLI values in our traces reflect per-fragment payload lengths (the attacker-visible observable).

Encryption note (PLI invariance). Payload bytes can optionally be AES-CTR protected; however, this does not change the PLI trace because AES-CTR is length-preserving (XOR), so encryption does not alter any fragment length. Our analysis uses only plaintext headers (e.g., PLI) and timing.

V. EXPERIMENT AND PRELIMINARY RESULTS

First, we played each of 10 videos (three runs each) on Ubuntu 24.04 under a fixed playback quality and resolution setting (720p). To minimize interference, we restricted other background traffic (e.g., VNC, SSH) during playback. For each run, we captured downlink ethernet packet (PCAP) for 180 seconds. Then, we replayed each PCAP file in our simulator to record the resulting PLI values and timestamps as CSV traces. We preserve original PCAP packet timestamps to retain idle periods and burstiness; each synthesized GEM is assigned `t_start_us` derived from the PCAP time base (μ s resolution). For similarity analysis, we bin PLI events into 100-ms bins, compute the cumulative sum, normalize by the final cumulative value, and compare traces using Pearson correlation and RMSE. These traces approximate the payload length observations available to a real same-splitter adversary and enable systematic evaluation of PLI based inference.

After producing the dataset, we visualized cumulative PLI-Time Plot based on our dataset (Publicly Available on our Github repository) [3]. We observed a clear correlation between each video and its cumulative PLI over time graph [3]; traces from the same video show highly consistent slopes across runs. (Our Github repository: /out_cumsum) We use run1,2 as a database and treat run3 as a query trace.

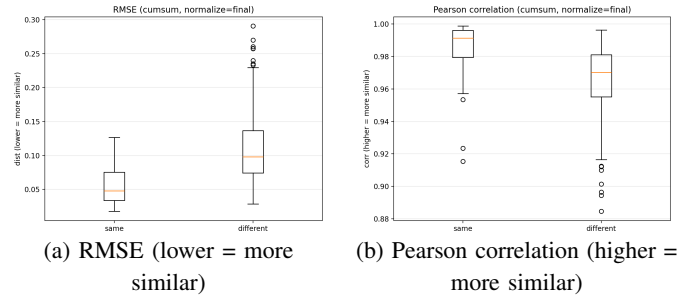


Fig. 1. Similarity distributions between same-video and different-video cumulative PLI traces. Same-video pairs tend to show higher correlation and smaller distance, with partial overlap due to ABR variability and buffering.

TABLE I
PER-QUERY TOP-1 PREDICTIONS. [3]

Query	Prediction	CORRECT	margin	r/RMSE
vid001/r3	vid001/r1	T	2	0.998/0.028
vid002/r3	vid002/r1	T	2	0.998/0.041
vid003/r3	vid003/r2	T	1	0.980/0.050
vid004/r3	vid004/r1	T	2	0.996/0.037
vid005/r3	vid005/r2	T	2	0.998/0.020
vid006/r3	vid006/r1	F	2	0.994/0.048
vid007/r3	vid007/r1	T	2	0.999/0.025
vid008/r3	vid008/r1	T	2	0.999/0.018
vid009/r3	vid009/r2	T	2	0.990/0.041
vid010/r3	vid010/r1	T	3	0.998/0.020

We further quantify similarity on the normalized cumulative PLI time series: same video pairs exhibit higher Pearson correlation (median $r = 0.991$ vs 0.970 , AUC=0.779) and lower RMSE (median 0.048 vs 0.098 , AUC=0.826). Using nearest-neighbor matching with rank-vote fusion of Pearson and RMSE (vote_pr), we achieve 90% Top-1 identification accuracy (9/10) when matching run3 queries against run1–run2 gallery traces.

VI. CONCLUSION AND DISCUSSION

Despite GEM Payload Encryption, GPON is still vulnerable to inferring streaming activity because PLI values in GEM headers can be exploited to infer a victim's video activity (e.g., Youtube). Potential mitigations include traffic shaping and obfuscating GEM metadata.

Future Work will be conducted in testbed with Real OLT, ONU and Optical Splitter. We will use FPGA device to capture GEM frames to parse downstream frames. We will expand to more videos and longer captures with changing various resolution, bitrate.

Extent to XGS-PON Networks. We assume that XGS-PON has same issue; XGEM Frame that XGS-PON uses has Payload Length Indicator Field like GEM. We will validate this by repeating our capture-and-matching evaluation on an XGS-PON testbed.

REFERENCES

- [1] ITU-T Recommendation G.984.3, *Gigabit-capable passive optical networks (GPON): Transmission convergence layer specification*.
- [2] Cisco, "Understand GPON Technology," Troubleshooting TechNotes. (Document ID: 216230)
- [3] Our Github Repository - <https://gitshare.me/repo/c8afd178-a072-4cee-a01c-8b3d12467a28>
- [4] S. Bae, et al., "Watching the Watchers: Practical Video Identification Attack in LTE Networks," in *31st USENIX Security Symposium (USENIX Security '22)*, 2022.
- [5] X. Zhang, et al., "Traffic spills the beans: A robust video identification attack against YouTube," *Computers & Security*, vol. 137, 2024, Art. 103623.
- [6] N. Hu et al., "Breaking Through the Diversity: Encrypted Video Identification Attack Based on QUIC Features," in *ESORICS 2024*, 2024.